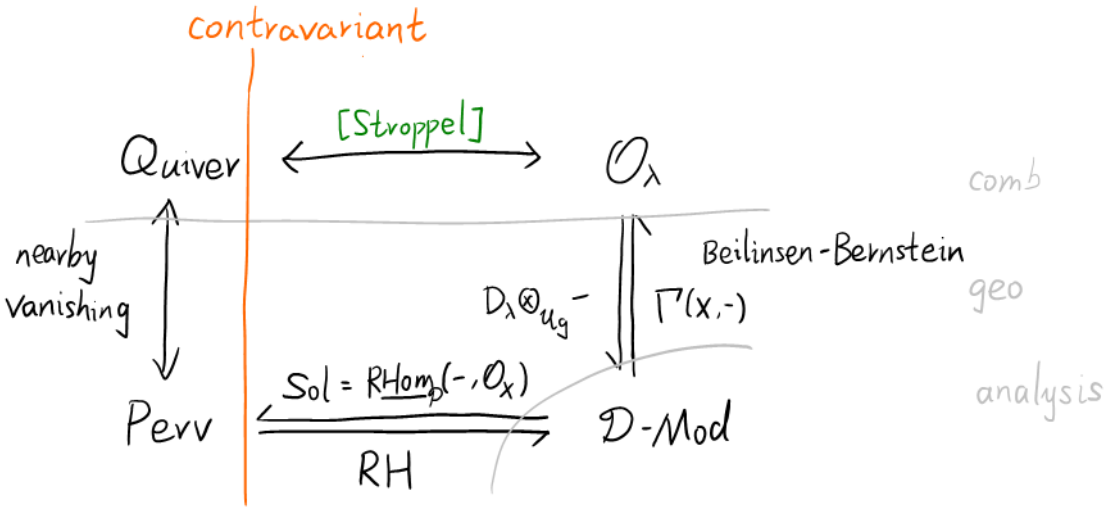
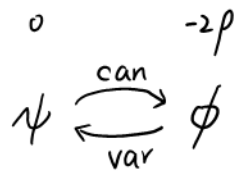
Eine Woche, ein Beispiel

11.10 5 indecomposible representations

This document is the continuation of [2013.11.26]. After the discussion with Renzhi Liang and Aaron, the last piece of the puzzle has been put together.

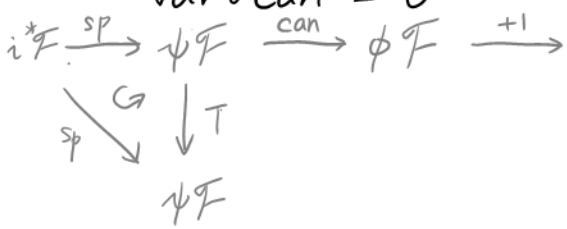
extra ref:
[Stroppel]: Category O: Quivers and endomorphism rings of projectives
https://www.math.uni-bonn.de/ag/stroppel/Quivers.pdf

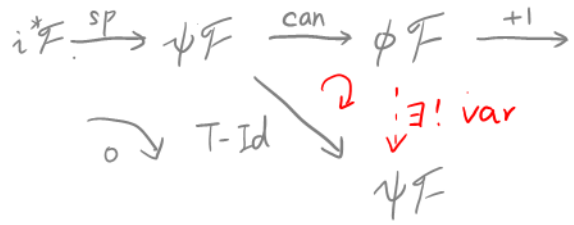


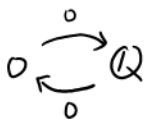
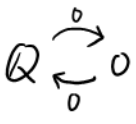
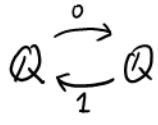
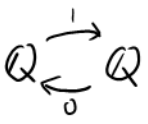
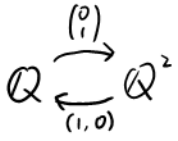


L : irreducible rep
M : Verma module
P : proj rep
I : inj rep

$\mathrm{var} \circ \mathrm{can} = 0$

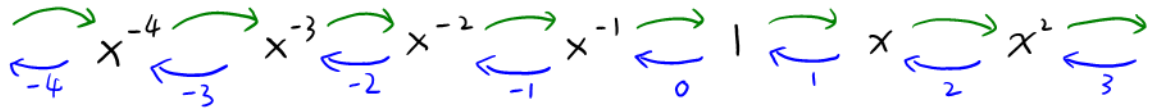




Quiver					
filtration					
Perv	$i_*\underline{\mathbb{Q}}_{\{\infty\}}$	$\underline{\mathbb{Q}}_{\mathbb{CP}^1}[1]$	$R_{j*}\underline{\mathbb{Q}}_{\mathbb{C}}[1]$	$j_!\underline{\mathbb{Q}}_{\mathbb{C}}[1]$	
alias	IC_0	IC_∞	$I(\psi)$	$P(\psi)$	$P(\phi) = I(\phi)$
$\mathcal{D}$-mod	A_1/A_1x	$A_1/A_1\partial$	$A_1/A_1x\partial$	$A_1/A_1\partial x$	$A_1/A_1x\partial x$
	$k[\partial]$	$k[x]$	$k[\partial, \partial^{-1}]$	$k[x, x^{-1}]$	
$\mathcal{O}_\lambda$	$L(-2\rho)$ $M(-2\rho)$ $M^*(-2\rho)$	$L(0)$	$M(0)$ $P(0)$	$M(0)^*$ $I(0)$	$P(-2\rho) = I(-2\rho)$
				dual	

Ex. the A_1-module structure of $k[x,x^{-1}]$

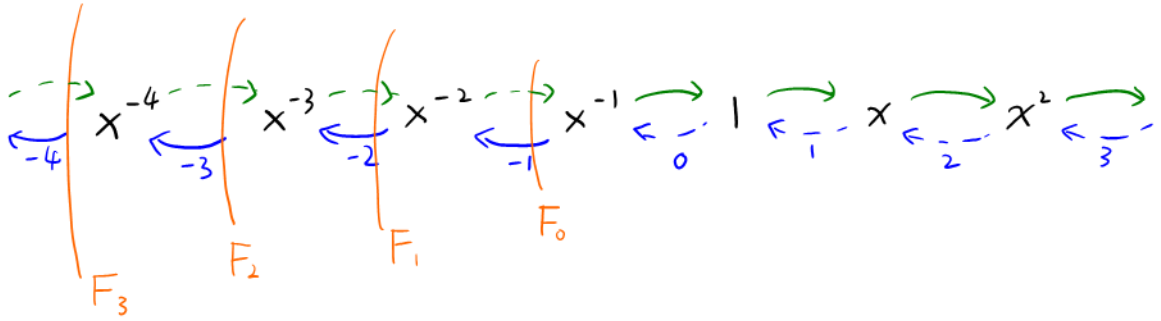
Basis with action:



$\rightarrow$: x-action

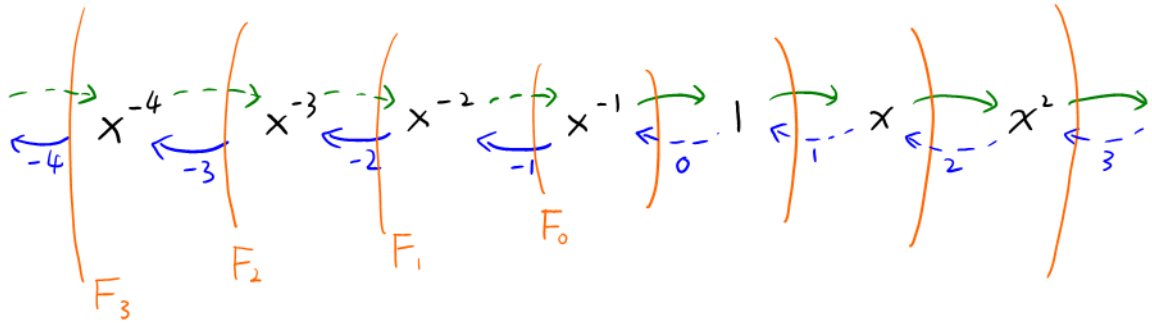
$\rightarrow$: ∂-action

Order filtration:



$\Rightarrow gr_{F^{ord}} k[x,x^{-1}] = k[x,\partial]/_{x\partial}$

Bernstein filtration:



$\Rightarrow gr_{F^B} k[x,x^{-1}] = k[x,\partial]/_{x\partial}$

Since $\partial x \cdot \frac{1}{x} = (x\partial + 1)\cdot\frac{1}{x} = 0$, we get

$$k[x, x^{-1}] \cong A_1/A_1\partial x \cong A_1/A_1(x\partial + 1)$$

We have a SES:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & k[x] & \longrightarrow & k[x,x^{-1}] & \longrightarrow & k[x,x^{-1}]/k[x] & \longrightarrow & 0 \\ & & \| & & \| & & \| & & \\ 0 & \longrightarrow & A_1/A_1\partial & \xrightarrow{\cdot x} & A_1/A_1\partial x & \longrightarrow & A_1/A_1 x & \longrightarrow & 0 \end{array}$$

$$0 \longleftarrow \underline{\mathbb{Q}}_{\mathbb{CP}^1}[1] \longleftarrow j_! \underline{\mathbb{Q}}_{\mathbb{C}}[1] \longleftarrow i_* \underline{\mathbb{Q}} \longleftarrow 0$$

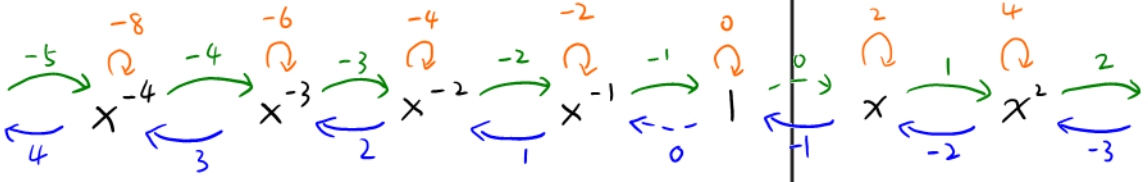
It does not split.

Restrict to $D_{\mathbb{C}^\times} = k[x, x^{-1}]\langle\partial\rangle$:

$$D_{\mathbb{C}^\times}/D_{\mathbb{C}^\times}\partial x = D_{\mathbb{C}^\times}/D_{\mathbb{C}^\times}\partial = K[x, x^{-1}]$$

Lie algebra action:

$M^*(0)$ | does not extend to $\mathbb{P}^1$



→ : ∂_e-action

→ : ∂_f-action

→ : ∂_h-action

$$\begin{cases} \partial_e = x^2\partial \\ \partial_f = -\partial \\ \partial_h = 2x\partial \end{cases}$$

Perverse sheaf interpolation:

$$R\Gamma(\mathbb{A}^1, \mathcal{P}) = R\mathrm{Hom}_{A_1}(k[x, x^{-1}], \mathcal{O}_{\mathbb{A}^1})$$

$$= R\mathrm{Hom}_{A_1}(A_1/A_1\partial x, A_1/A_1\partial)$$

$$= R\mathrm{Hom}_{A_1}([A_1 \xrightarrow{\partial x} A_1], A_1/A_1\partial)$$

$$= [A_1/A_1\partial \xleftarrow{\partial x\cdot} A_1/A_1\partial]$$

$$= [k[x] \xleftarrow{\partial x\cdot} k[x]]$$

$$= 0$$

$$R\Gamma(\mathbb{G}_m, \mathcal{P}) = [k[x, x^{-1}] \xleftarrow{\partial x\cdot} k[x, x^{-1}]]$$

$$= k \oplus k[-1]$$

For $\beta \in \mathrm{MaxSpec}\, k[x, x^{-1}]$,

$$\mathcal{P}_\beta \neq [k[x, x^{-1}]_\beta \xleftarrow{\partial x\cdot} k[x, x^{-1}]_\beta]$$

$\mathcal{P}_\beta = k$?